BY ORDER OF THE
SECRETARY OF THE AIR FORCE

DEPARTMENT OF THE AIR FORCE
MANUAL 91-222

6 MARCH 2025

Safety

SPACE SAFETY INVESTIGATIONS AND REPORTS

COMPLIANCE WITH THIS PUBLICATION IS MANDATORY

ACCESSIBILITY: Publications and forms are available for downloading or ordering on the e-Publishing website at www.e-Publishing.af.mil.

RELEASABILITY: There are no releasability restrictions on this publication.

OPR: AFSEC/SES

Certified by: AF/SE
(Colonel Michael A. Thomas)

Supersedes: AFMAN 91-222, 17 June 2019

Pages: 28

This publication implements Department of the Air Force Instruction (DAFI) 91-204, *Safety Investigations and Reports*. It provides additional guidance and procedures for safety investigations and reports involving space systems and/or their unique support equipment and systems. It applies to the entire Department of the Air Force (DAF), including DAF civilian employees and uniformed members of the United States Air Force and the United States Space Force, the Air Force Reserve, the Air National Guard, and the Civil Air Patrol when conducting missions as the official Air Force Auxiliary, as well as those with a contractual obligation to abide by the terms of DAF issuances, except where noted otherwise. Ensure all records generated as a result of processes prescribed in this publication adhere to Air Force Instruction (AFI) 33-322, *Records Management and Information Governance Program*, and are disposed in accordance with the Air Force Records Disposition Schedule, which is located in the Air Force Records Information Management System. Refer recommended changes and questions about this publication to the office of primary responsibility (OPR) using the DAF Form 847, *Recommendation for Change of Publication*; route DAF Forms 847 from the field through the appropriate functional chain of command. This publication may be supplemented at any level, but all direct Supplements must be routed to the OPR of this publication for coordination prior to certification and approval. The authorities to waive wing, unit, or delta level requirements in this publication are identified with a tier ("T-0, T-1, T-2, T-3") number following the compliance statement. See Department of the Air Force Manual (DAFMAN) 90-161, *Publishing Processes and Procedures*, for a description of the authorities associated with the tier numbers. Submit requests for waivers through the chain of command to the appropriate tier waiver approval authority, or alternately, to the requestor's command or director (or equivalent), for non-tiered compliance items and provide a copy of the

approved waivers to AFSEC/SES. This instruction requires the collection and or maintenance of information protected by the Privacy Act of 1974 in accordance with DoDI 5400.11, *DoD Privacy and Civil Liberties Programs*.

SUMMARY OF CHANGES

This publication has been rewritten and should be completely reviewed. Major changes include adding United States Space Force language where appropriate, clarifying research, development, test, and evaluation reporting requirements, clarifying on-orbit anomaly reporting requirements, and adding investigation and safety report procedures.

Chapter 1—OVERVIEW **4**

1.1. Overview. 4

1.2. Safety Investigation Board (SIB) Support. 4

Chapter 2—ROLES AND RESPONSIBILITIES **5**

2.1. General. 5

2.2. Convening Authority. 5

2.3. Space Launch Delta Commanders and Other DAF Range Commanders. 5

2.4. Unit Commanders or Equivalent. 5

2.5. Director/Chief of Safety/Mission Safety. 6

Chapter 3—GENERAL INFORMATION **7**

3.1. Additional Reporting Requirements. 7

3.2. Unintentional Release of Radioactive Material. 7

3.3. Class E Mishaps. 7

3.4. Exception: 7

3.5. Orbital Anomaly Resolution. 7

3.6. Operational Spacecraft Space Mishap Classification. 7

3.7. Research Development Test and Evaluation (RDT&E) Space Mishap Classification. 8

3.8. Space Safety Mishap Costs. 8

3.9. On-orbit Event Reporting. 8

3.10. Hazard Reporting. 8

Chapter 4—INVESTIGATION MEMBERSHIP **10**

4.1. General Information. 10

4.2. Interim Safety Board. 10

4.3. Interim Safety Board (ISB) Evidence Collection and Preservation. 10
4.4. SIB Membership. 10
4.5. SIB Member Requirements and Duties. 11
4.6. DAF Participation in Non-DAF investigation. 13
Chapter 5—INVESTIGATION PROCEDURES AND TECHNIQUES 15
5.1. Procedures and Techniques. 15
5.2. Privileged Safety Information. 15
5.3. SIB Day One Briefing. 15
5.4. Non-Disclosure Agreements (NDA). 15
5.5. SIB Evidence Collection and Analysis. 15
5.6. Transferring and Disposing of Evidence. 16
5.7. Temporary De-Convening. 16
5.8. SIB Conclusion. 17
Chapter 6—SAFETY REPORTS 18
6.1. Safety Reports. 18
6.2. Final Message. 18
6.3. AFSAS Report Requirements. 18
6.4. DoD Human Factors Analysis and Classification System (HFACS). 19
6.5. Managing Exhibits. 19
6.6. Exhibit Waivers. 22
6.7. Safety Report Authentication. 22
6.8. Classified Reporting. 22
Attachment 1—GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION 23
Attachment 2—ORGANIZATION CONTACT INFORMATION 28

Chapter 1

OVERVIEW

1.1. Overview.

1.1.1. This DAFMAN, in conjunction with DAFI 91-204, provides space discipline-specific guidance for investigating and reporting DAF safety events. In this publication, the term "command" includes Field Commands (FLDCOMs), Major Commands (MAJCOMs), and MAJCOM-like entities (higher headquarters), including the Air National Guard, Field Operating Agencies (FOAs), and Direct Reporting Units (DRUs).

1.1.2. For the purposes of this publication, the term "Safety Investigation Board (SIB)" includes any assembly of safety investigators or a single investigating officer (SIO).

1.2. Safety Investigation Board (SIB) Support. SIBs should contact their Convening Authority safety staff for investigative support. Contact the Air Force Safety Center Space Safety Division (AFSEC/SES) staff when the Convening Authority safety staff cannot provide the required support in a timely manner.

Chapter 2

ROLES AND RESPONSIBILITIES

2.1. General. The DAF generally assigns investigative responsibilities to the command experiencing operational loss of an assigned/owned asset, or the procuring command whose system or service resulted in unintended damage or injury. DAF organizations that have procurement, operational, or sustainment responsibility must report all safety events involving space systems to their safety office immediately following notification. Permanent capability degradation or loss of an orbital Department of Defense (DoD) space vehicle prior to planned end-of-life constitutes damage.

2.2. Convening Authority. In addition to the DAFI 91-204 Convening Authority requirements, the Convening Authority at all levels will:

2.2.1. Investigate safety events per Safety Investigation Board (SIB) membership requirements. Convening Authorities may tailor investigation membership to suit unique space-specific circumstances. (see **paragraph 4.4**.)

2.2.2. For non-Federal Aviation Administration (FAA)-licensed launch vehicles, including upper stages, assign ownership of the mishap to the command acquiring the launch vehicle or launch service unless an agreement specifies ownership transfer to another agency.

2.2.3. Ensure SIB members meet the minimum training, qualification, and grade requirements. (see **paragraph 4.5**.)

2.3. Space Launch Delta Commanders and Other DAF Range Commanders.

2.3.1. Before each United States government operation (e.g., launch, fly-back, reentry), commanders of DAF ranges will:

2.3.1.1. Identify and notify organizations that have investigative responsibility in the event of a mishap.

2.3.1.2. Consider identifying an Interim Safety Board (ISB) for high-risk operational activities with significant mishap potential.

2.3.2. Notify the FAA Office of Commercial Space Transportation (FAA/AST) when a mishap occurs involving a licensed commercial space system at a DAF range.

2.3.3. Contact AFSEC/SES and MAJCOM/FLDCOM/DRU safety office regarding established Memorandums of Agreement/Memorandums of Understanding (MOAs/MOUs) between the DAF, FAA, and National Transportation Safety Board (NTSB) to determine appropriate investigative actions.

2.4. Unit Commanders or Equivalent.

2.4.1. Ensure MOAs/MOUs define ownership and responsible parties for investigating space mishaps for Research and Development Test and Evaluation (RDT&E) programs with participation from multiple government mission partners.

2.4.2. Ensure the unit space safety manager is notified of on-orbit anomalies lasting more than 5 days. (see **paragraph 3.5**.)

2.5. Director/Chief of Safety/Mission Safety. In addition to the investigation requirements in DAFI 91-204, the Director/Chief of Safety/Mission Safety or equivalent will:

2.5.1. Ensure an adequate number of potential investigators are appropriately trained and available to conduct space safety investigations.

2.5.2. Advise the Convening Authority on space-specific SIB membership.

2.5.3. Ensure safety staffs accomplish SIB appointment orders. (see **paragraph 4.4.2**.)

2.5.4. Ensure safety staffs set up Air Force Safety Automated System (AFSAS) accounts for each SIB member/investigator. (see **paragraph 4.4.3**.)

2.5.5. Provide the SIB with contact information for ISB.

2.5.6. Provide the SIB with a point of contact from the Convening Authority's safety staff. The point of contact aids and advises in all aspects of the investigation.

2.5.7. Monitor the progress of open investigations.

2.5.8. Review extension requests and obtain Convening Authority approval for extensions when appropriate.

2.5.9. Assist SIBs in determining and contacting the OPR and office of collateral responsibility for recommendations and Other Recommendations of Significance (ORS).

Chapter 3

GENERAL INFORMATION

3.1. Additional Reporting Requirements. In addition to the event reporting requirements in DAFI 91-204, unit safety staffs will report the following unique space events:

3.1.1. Damage to DAF space systems during pre-launch or launch on a commercially procured FAA-licensed launch system or service. **(T-1)**

3.1.2. United States government fly-back system that results in damage or injury or a commercial fly-back system that causes damage to DAF resources. **(T-1)**

3.1.3. Damage or injury resulting from the reentry of a DAF-owned space system, such as a spacecraft, rocket body, or debris. **(T-1)**

3.1.4. Damage to unique DAF space system ground support equipment. **(T-1)**

3.2. Unintentional Release of Radioactive Material. See Department of the Air Force Manual (DAFMAN) 40-201, *Radioactive Materials (RAM) Management*, for situations involving the unintentional release of radioactive material.

3.3. Class E Mishaps. There are no Class E events that require mandatory reporting.

3.4. Exception: Safety staffs will not report normal residual damage that results from a space launch or pre-defined emergency response actions. Additionally, do not report damage to space systems that fail to operate in environments exceeding their design requirements. For research development test and evaluation (RDT&E) exceptions (see **paragraph 3.7**.).

3.5. Orbital Anomaly Resolution. Orbital mishap declarations are unique in that anomaly resolution actions are typically accomplished prior to a mishap being declared.

3.5.1. The operational unit, as the key player responsible for the spacecraft, plays a crucial role in the reporting process. They will notify the unit safety office of any mission-impacting event, unplanned loss of operational capability, or unresolved orbital anomaly affecting on-orbit spacecraft that extends past five days from the event's first detection. **(T-2)** The unit safety office will keep the FLDCOM safety office abreast of significant anomaly resolution activity until the anomaly is resolved or a mishap is declared. **(T-2**)

3.5.2. Convening Authorities, at their discretion, can support a space safety investigation using a programs operational anomaly resolution process for analyzing technical data and other fact-based information.

3.6. Operational Spacecraft Space Mishap Classification.

3.6.1. In addition to total direct mishap cost on-orbit space mishaps are also classified by severity of mission loss or degradation.

3.6.2. Operational on-orbit space systems which have not yet reached the end of their design life will have mishaps classified based on the impact to mission capability. **(T-1)**

3.6.2.1. This does not include non-operational systems (i.e., systems that did not undergo a formal operational acceptance process).

3.6.2.2. A program's Capability Development Document or other operational capability requirements document defines the spacecraft's mission capability. The document should specify primary, secondary, and tertiary mission capabilities.

3.6.3. Units will report events involving on-orbit space systems operating beyond their design life as hazards (see **paragraph 3.10**). **(T-1**)

3.6.4. Units will report and investigate unplanned loss or degradation of operational mission capability as safety events described below:

3.6.4.1. Classify permanent loss of primary mission capability as a Class A mishap. **(T-1)**

3.6.4.2. Classify permanent degradation of primary or secondary mission capability or the permanent loss of secondary mission capability as a Class B mishap. **(T-1)**

3.6.4.3. Classify permanent loss or degradation of tertiary mission capability as a Class C mishap. **(T-1)**

3.7. Research Development Test and Evaluation (RDT&E) Space Mishap Classification. RDT&E activities are by their very nature high risk and not operational. For RDT&E failures, safety staffs will report space safety events for unexpected outcomes resulting in damage that are not addressed in the test approval documentation. **(T-2)** A viable RDT&E test plan defines credible objectives and expected outcomes. Damage or loss of mission experienced during testing not listed in the test plan or in formal risk acceptance documentation as an expected or credible scenario is investigated as a mishap per Department of Defense Instruction (DoDI) 6055.07, *Mishap Notification, Investigation, Reporting & Record Keeping*. **(T-0)**

3.8. Space Safety Mishap Costs. The DoD requires determining the total direct mishap cost to provide a factual basis for allocating resources supporting DoD mishap prevention programs. **(T-0)** Determine mishap costs per DAFI 91-204. **(T-1)**

3.8.1. For orbital mishaps, the orbital segment is a unique case as actual hardware cannot be accessed or repaired (beyond software modification); thus, special considerations will be taken to determine appropriate mishap costs. **(T-2)**

3.8.2. For permanent loss of primary mission, use the cost of the spacecraft at launch. **(T-1)** The SIB should work with the system program office to determine the cost at the time of launch.

3.8.3. For loss of secondary or tertiary mission systems, determine the portion of the spacecraft cost at the time of launch attributable to the failed mission system(s) and use that as the mishap cost. **(T-1)**

3.8.4. For permanent partial mission degradation, including mishaps, which negatively affect mission service life, the SIB must consult with the systems program office and operator(s) to determine the mission degradation percentage and multiply it by the spacecraft cost at launch to determine the cost of the mishap. **(T-1)**

3.9. On-orbit Event Reporting. MAJCOM/FLDCOM/DRU safety staffs will monitor the status of significant on-orbit anomalies until resolution or enough information exists to classify the anomaly as a reportable event. **(T-1)**

3.10. Hazard Reporting. At a minimum, units will report the following as a hazard in AFSAS:

3.10.1. Any unauthorized creation of orbital debris, 5mm or larger, that involves DAF space systems. **(T-1)**

3.10.2. Loss or permanent degradation to an operational spacecraft performing missions beyond its expected design life. **(T-1)**

3.10.3. Anomaly preventing adherence to the spacecraft end of life plan or the United States Government Orbital Debris Mitigation Standard Practices (ODMSP). **(T-1)**

3.10.4. The space systems acquisition office will report inactive (e.g., rocket bodies or inactive spacecraft) on-orbit creation of debris. **(T-1)**

Chapter 4

INVESTIGATION MEMBERSHIP

4.1. General Information. Conduct investigations to determine the cause of an event through the deliberative process of evidence gathering and root cause analysis to provide sound recommendations to prevent future mishaps.

4.2. Interim Safety Board. The purpose of the ISB is to gather and preserve evidence after a mishap and protect privileged safety information, in accordance with DAFI 91-204. **(T-1)** ISB actions are limited to identifying evidence, taking control or possession of evidence, and preservation until transfer to the SIB. The ISB will not investigate or determine factors or causes of a mishap. **(T-1)**

4.2.1. An ISB may be required in more than one location. In this situation, the Convening Authority will designate a lead ISB President to coordinate and control the activities at all locations. **(T-2)**

4.2.2. The ISB will impound and control processing of all mission recorded data (paper, electronic, imaging, etc.) at the range or satellite control facility. Upon arrival, authorize the release and transfer of impounded data to the SIB president. **(T-2)**

4.2.3. Depending on the mishap, an ISB may consist of one individual or several personnel. ISB members will not be from the mishap wing/delta or equivalent organization. **(T-2)** They will not have been directly involved in the mishap or involved in the immediate supervision or execution of the activity that led up to the mishap. **(T-2)**

4.2.4. ISB members are selected based on qualifications, grade, and training.

4.2.4.1. The ISB President must be a graduate of the Board President Course. **(T-1)**

4.2.4.2. The ISB IO will be a graduate of a DAF mishap investigation course: the Space Mishap Investigation Course (SMIC), the Aircraft Mishap Investigation Course (AMIC), or the Mishap Investigation Non-Aviation Course (MINA). **(T-1)**

4.3. Interim Safety Board (ISB) Evidence Collection and Preservation. Evidence collected by the ISB lays the groundwork necessary for a successful investigation by the SIB. The ISB assembles evidence from multiple sources and locations; concentrating only on the mishap site may delay the collection of perishable evidence.

4.3.1. ISBs will collect recorded information, electronic data, or telemetry from spacecrafts, ground stations and other recorded space systems data that provides valuable evidence for mishap investigations. **(T-1)**

4.3.2. ISBs should consult with the operations officer and the chief engineer as appropriate for recorded information, electronic data, and telemetry that may be relevant to the mishap investigation. The information, data, or telemetry may be classified, and special procedures may be required to download, transfer, and preserve this evidence. If required, contact the local Special Security Officer to coordinate a transfer. **(T-2)**

4.4. SIB Membership. The Convening Authority may tailor SIB member personnel requirements for Primary, Secondary, and support members as appropriate for space mishaps.

4.4.1. At a minimum, the Convening Authority of a space safety investigation must appoint an Investigating Officer (IO). **(T-1)** An IO led investigation should consider a recorder to assist in investigation administrative duties.

4.4.2. Safety staffs must ensure orders appointing the SIB are accomplished. **(T-1)** The orders must include all SIB primary members, the recorder, and may include other secondary members. **(T-1)** The orders must contain the individual's position, full name, rank/grade, organization, and assigned duty location. **(T-1)** Accomplishment of orders within AFSAS is optional using the automate preparation and archiving tool or upload a copy to the exhibits group appointment orders field.

4.4.3. Each primary SIB member must have an AFSAS account. **(T-2)** At a minimum, the following roles should be assigned:

4.4.3.1. Data Extraction Access: Events

4.4.3.2. Data Extraction Access: Recommendations

4.4.3.3. Event: Data Viewer

4.4.3.4. Event: Investigator

4.4.3.5. Event: Message View Only

4.4.3.6. Recommendation: View Only

4.4.3.7. Workspace: United States Air/Space Force

4.4.4. Primary Members. Primary members determine the SIBs results, including factors, findings, causes, and recommendations. Primary members include a SIB Board President (BP), IO, and Air Force Safety Center (AFSEC) representative, if applicable. When the circumstances of the mishap require additional expertise, individuals identified in **paragraph 4.5** may be added as primary members. Convening Authorities will ensure primary members are DoD personnel. **(T-1)** Primary members sign the investigation authentication page and may submit Minority Reports if they do not concur with the SIBs results in whole or in part.

4.4.5. Secondary members. Secondary members provide expertise or assistance to the SIB. Secondary members may participate in all SIB deliberations but will not sign the authentication page or submit a Minority Report. The SIB president determines the extent of the secondary member's involvement in the SIB.

4.4.6. Support members. Support members are observers or other personnel who assist the SIB. Support members can come from depots, industry, civilian organizations, Convening Authority safety staffs, other government organizations, and laboratories. The SIB president decides how much support members can be involved in the SIB. Support members do not need to be DoD personnel.

4.5. SIB Member Requirements and Duties.

4.5.1. SIB Board President. The BP is responsible for all activities of the SIB, is the deciding vote if a disagreement within the SIB occurs and is the final resolution authority who reports directly to the Convening Authority. The BP coordinates site access requirements directly with the incident commander or local site commander. The BP must be a graduate of the Safety and

Accident Investigation Board President Course (BPC). **(T-1)** The BP will be from a squadron/organization different from the one that experienced the mishap. **(T-1)**

4.5.2. Investigating Officer (IO). The IO is responsible for the daily SIB activities, the investigation, and the preparation of reports and messages. The IO directs and coordinates the activities of other board members and works with the AFSEC/SES Representative (if assigned) to manage the investigation. The IO will be a graduate of the Space Mishap Investigation Course (SMIC), implemented on 1 January 2023. SMIC has replaced the Aviation Mishap Course (AMIC) and the Mishap Investigation Non-Aviation (MINA) Course. An IO who is a graduate of AMIC or MINA prior to 1 January 2023, is qualified to perform the duties of an IO. **(T-1)** For Class A and B mishaps with no BP, the IO must always be equal to or higher in rank than the highest-ranking individual involved in the mishap. **(T-1)** The IO will be from a squadron/organization different from the one that experienced the mishap. **(T-1)**

4.5.3. AFSEC Representative. AFSEC/SES may provide a representative or advisor to the SIB for Class A and complex Class B mishaps, as determined by AFSEC/SES in consultation with the Convening Authority safety staff.

4.5.3.1. The representative acts as the investigation process expert to guide SIB procedures and analysis efforts, conducts refresher training on SIB procedures, coordinates technical assistance resources, and is the single point of contact with AFSEC for the investigation.

4.5.3.2. If the representative supports the SIB remotely, the representative is a consultant to the SIB and not a primary member.

4.5.4. Recorder. The recorder will be an officer/non-commissioned officer/civilian (O-1, E-4, GS-5, or higher) familiar with administrative procedures and experienced in the use of typical office computer software or an inexperienced graduate of SMIC with the intent to gain knowledge on the investigative processes. **(T-2)** They will manage the work center, control access, security, report preparation, and distribution. **(T-2)**

4.5.5. MAJCOM/FLDCOM/DRU Representative. The representative acts as the investigation process expert to guide SIB procedures and analysis efforts. The representative is responsible for providing basic information on the operational and organizational details and practices to help the SIB determine who to interview, mishap organization hierarchy, etc. This individual will be considered an expert on local procedures, local command relationships, organizational structure, and unit-specific qualities. **(T-2)**

4.5.6. Space Operations Representative. The space operations representative analyzes operations factors, including but not limited to, qualifications, proficiency, training, communications, crew actions throughout the mishap sequence, mission-specific concerns, performance data, operations-related publications, and crew stressors. The space operations representative must be qualified or previously qualified in the use of a space system like that involved in the mishap. **(T-2)**

4.5.7. Space Acquisition Representative. Representatives act as the technical expert on all phases of the acquisition life cycle, including, but not limited to, development, production and manufacturing, deployment, acquisition logistics, configuration and data management, contracting, engineering, program control, system safety, quality assurance, and test and evaluation.

4.5.8. Weather or Space Environmental Representative. Convening Authorities can utilize a space weather representative if the space environment is suspected or a known factor in the safety mishap. A space weather representative is a subject matter expert on analyzing and forecasting meteorological, solar, astrophysical, geophysical, and atmospheric weather conditions.

4.5.9. Medical Officer. Officers analyze and evaluate medical data, histories, records and pathology reports. Furthermore, they determine the cause(s) and mechanisms of injuries or death. The medical member must be a Flight Surgeon and a graduate of the Aircraft Mishap Investigation and Prevention Course. **(T-1)** If no flight surgeon is assigned to the SIB, a flight surgeon from the mishap wing/delta or equivalent may review medical and dental records for the SIB. If the flight surgeon identifies areas that may have contributed to the mishap, the Convening Authority will assign a flight surgeon to the SIB. **(T-2)**

4.5.10. Cyber Representative. Representatives are subject matter experts on the cyber domain associated with the mishap. The cyber representative will analyze and determine if software (or hardware) contributed to the mishap.

4.5.11. Many different types of subject matter experts are available to SIBs. These experts and advisors can come from depots, industry, civilian organizations, other government organizations, and laboratories. Typically, government personnel are primary and secondary members and contractors are supporting members however non-government personnel can be secondary members.

4.6. DAF Participation in Non-DAF investigation. The DAF is always responsible for determining the cause of DAF mishaps. In certain situations, the DAF may accept the results of a non-DAF investigation to satisfy DAF mishap prevention requirements as directed by the Convening Authority. Outside agencies such as the FAA, National Aeronautics and Space Administration (NASA), and/or NTSB may allow DAF personnel to participate as official observers to investigations led by these agencies. Reference Air Force Pamphlet (AFPAM) 91-206, *Participation in Military or Civil Aircraft or Space Safety Investigations*, for further guidance on participation in interagency investigations. The DAF may utilize these investigations to limit duplicative investigations. However, the DAF always retains the right to conduct a separate safety investigation.

4.6.1. DAF observers on non-DAF investigation will:

4.6.1.1. Contact the appropriate command safety office or AFSEC/SES for guidance on current agreements with the FAA, NASA, NTSB, or the responsible commercial entity. **(T-1)**

4.6.1.2. Protect all investigation data, information, documents, or other material as prescribed by the investigating agency. **(T-0)**

4.6.1.3. Not disclose investigation information in any manner without the written consent of the Investigator-in-Charge. **(T-0)**

4.6.1.4. Refer any Freedom of Information Act request, subpoena, or other request for such information to the FAA, NASA, and/or NTSB for resolution. **(T-0)**

4.6.1.5. Not offer promise of confidentiality. **(T-0)** Observers are not formal DAF safety investigators.

4.6.1.6. Not observe multiple investigations simultaneously nor may they participate in any other investigations while still acting in an observer status. **(T-1)** Observers may transition from an DAF observer to a DAF Safety Investigation Board to provide the SIB with continuity. However, once transferred, they may no longer serve as an DAF observer, or transfer back to observer status. **(T-1)**

4.6.2. Use of non-DAF investigations.

4.6.2.1. The DAF observer reports to the Convening Authority on the investigation's adequacy and the conclusions' validity. The observer also advises whether the conclusions satisfy DAF's mishap prevention needs. If the report fulfills the requirements of a DAF final report, the Convening Authority may use the non-DAF investigation report as the official report.

4.6.2.2. Based on the DAF observer's assessment of the adequacy of the non-DAF investigation report, the DAF Convening Authority may accept it with a cover letter from the DAF observer, stating acceptance of findings and submit it as the DAF final report within AFSAS. Alternately, the Convening Authority may appoint an investigation officer to evaluate the report and deliver a DAF mishap report. In this case, the DAF report will include the non-DAF report. **(T-1)**

4.6.2.3. If the DAF Convening Authority determines that the investigation does not satisfy the DAF's mishap prevention needs, the Convening Authority will conduct a DAF investigation within the terms of applicable contracts and MOAs/MOUs. **(T-0)**

Chapter 5

INVESTIGATION PROCEDURES AND TECHNIQUES

5.1. Procedures and Techniques. This chapter augments guidance in DAFI 91-204, providing additional procedures and techniques uniquely applicable to space safety investigations.

5.2. Privileged Safety Information. Before the ISB handover briefing, the SIB IO or AFSEC representative will brief SIB members on the definition and protection of privileged safety information. **(T-0)** Investigators will accomplish safety privilege training via myLearning—Protecting Privileged Safety Information. All SIB members will sign the "SIB Member Guidance and Non-Disclosure Agreement (NDA)" after receiving the privileged safety information training. **(T-1)** The SIB will use the template located in the SIB Go Package. **(T-1)**

5.3. SIB Day One Briefing. After the ISB handover briefing, the SIB IO or AFSEC representative will provide the "SIB Day One" briefing to the SIB members. **(T-2)** The briefing overviews SIB member roles and responsibilities, the investigation process, and reporting timelines. The SIB IO or AFSEC representative ensures that SIB primary members review their associated checklists in the SIB Go Package.

5.4. Non-Disclosure Agreements (NDA). During the investigation, SIBs will ensure all other personnel exposed to privileged safety information sign a "Non-Disclosure Agreement – Safety Investigation DoD Personnel" or "Non-Disclosure Agreement-Safety Investigation-Contractor Rep" as applicable. **(T-0)** These personnel include but is not limited to interview transcribers, communication squadron personnel aiding the SIB, and individuals providing technical expertise. The SIB will use the templates located in the SIB Go Package. **(T-1)**

5.5. SIB Evidence Collection and Analysis. SIBs use all available information to investigate mishaps. SIBs will collect, review, and analyze all witness testimony, materiel analysis, historical documents, data, publications, etc., to determine the mishap's root cause(s). **(T-1)** Mishap evidence forms the basis of the SIB's safety report. SIBs ensure evidence is controlled at the appropriate classification level. Guidance on SIB evidence collection and analysis is provided in DAFI 91-204, SIBs will:

5.5.1. Collect and review records for all involved personnel (e.g., medical, training), space systems, and equipment as needed. **(T-1)**

5.5.1.1. Analyze records to identify any abnormalities that may have contributed to the mishap. **(T-1)**

5.5.1.2. As the investigation progresses, continue to collect and analyze records of other personnel or equipment that may have contributed to the mishap, as appropriate. **(T-1)**

5.5.2. Collect and analyze all digital and video images. **(T-1)** Images captured by the ISB, first responders, or other witnesses may be crucial in identifying mishap cause(s) or eliminating damage caused by recovery actions.

5.5.2.1. Maintain a list and description of photographs or videos taken. **(T-2)** This listing must include, at a minimum, the location and name of the items recorded. **(T-2)**

5.5.2.2. Photograph/video parts or equipment before disassembling or analyzing. **(T-2)**

5.5.2.3. Do not upload classified photographs or videos to AFSAS. **(T-1)**

5.5.3. Collect and analyze telemetry data for on-orbit events. **(T-1)** This data may aid the SIB in determining when events occurred during the mishap sequence.

5.5.3.1. Data sources may include downlink (mission and status of health, or information from various subsystems, including the payload) telemetry data and space weather data. If applicable, SIBs will collect space weather from similar spacecraft transiting the same region. **(T-1)** Additionally, contact the 557th Weather Wing/2d Weather Squadron/Space Weather Operations Center (SpaceWOC) for space weather data information**. (T-1)** See DAFI 91-204 for guidance on shipping recorders.

5.5.3.2. Collect, review, and analyze the command plan(s) that were uplinked prior to any on-orbit anomaly. **(T-1)** Investigators will determine if the operators sent command(s) out of sequence or if the command(s) were correct for the situation and evaluate if the resulting telemetry was correct or if it indicated an issue. **(T-1)** A copy of the vehicle's "Command and Telemetry Handbook" may be necessary to understand the command structure.

5.6. Transferring and Disposing of Evidence. The SIB will provide all non-privileged evidence to the Accident Investigation Board (AIB) or other legal board, the host installation Staff Judge Advocate, or return to the owning organization, per DAFI 91-204. **(T-1)**

5.6.1. If the mishap organization requests the return of records or equipment to complete the mission, use guidance in DAFI 91-204. If the mishap organization requests the return of records for other reasons (e.g., medical care, training), and if the BP determines it is appropriate, the SIB will retain the original documents and provide copies to the requesting organization. **(T-1)** The SIB will document the return of records or equipment using the "Evidence Transfer Memo – No AIB" signed by the BP and the owning organization commander. **(T-1)** The SIB must use the template located in the SIB Go Package (Evidence Transfer Flow Chart). **(T-1)**

5.6.2. SIBs may provide non-privileged information to a legal investigation as it becomes available, but not to the detriment of the SIB. This information includes but is not limited to maintenance records, toxicological results, non-privileged technical analysis reports, non-privileged photographs, and medical records. When the SIB transfers evidence to the legal investigation, whether before or after the completion of the SIB, the transfer must be documented using the "Evidence Transfer Memo." The Evidence Transfer Memo must be signed by a representative of the SIB and the legal investigation Board President or designated representative. **(T-1)** The SIB must use the template located in the SIB Go Package. **(T-1)** The BP will release witnesses, participants, and interviewees (e.g., mishap operator, mishap maintainer, mishap engineer) to a legal investigation when determined the SIB no longer needs them. **(T-1)**

5.6.3. SIBs will not release analysis, findings, causes recommendations, or references to privileged witness statements. **(T-0)** Additionally, do not release recordings of simulated, computer-generated, animated, or reenacted portions of the mishap if they involve privileged analysis (such as modified photos, diagrams, or charts showing SIB deliberations). **(T-0)**

5.7. Temporary De-Convening. SIBs may request to de-convene the investigation for one of the following reasons: waiting to out-brief the Convening Authority, waiting on technical analysis or wreckage recovery, or criminal acts are suspected. The Convening Authority must approve the SIB's request to de-convene. **(T-2)**

5.7.1. The SIB must upload into AFSAS all available exhibits and draft the narrative with all available information before de-convening while waiting on technical analysis or wreckage recovery. **(T-2)** The SIB must include placeholders in the narrative for analysis results or wreckage recovery. **(T-2)** The SIB must develop a plan that allows enough time to finish the investigation and complete the safety report and briefing. **(T-2)** The SIB remains the members' primary duty while de-convened. They are not relieved of their investigative responsibilities and must continue to monitor investigation-related activities. **(T-2)** The SIB must reconvene in person or via video or teleconference to complete the investigation. **(T-1)**

5.7.2. Before de-convening while waiting to out-brief the Convening Authority, the SIB must upload all exhibits and complete the required data fields in AFSAS. **(T-1)** Prepare the final message in AFSAS, but do not select "Submit for Release" to the Convening Authority. The SIB will complete the Mishap Quality Control Checklist in the "Quality Control & Memorandum of Final Evaluation (QC & MOFE)" folder of the SIB Go Package. **(T-1)** The SIB must notify the Convening Authority safety office once all exhibit requirements, the final message, and the Mishap Quality Control Checklist are complete. **(T-2)** The Convening Authority safety office will review all products for quality control. (**T-1)**

5.8. SIB Conclusion. After concluding the investigation and releasing the SIB, the Convening Authority safety staff, and AFSEC/SES will accomplish the following actions:

5.8.1. Transfer or dispose of all non-privileged evidence per DAFI 91-204. **(T-1)** SIBs will use the "Evidence Transfer Memo – No Accident Investigation Board (No AIB)" or the "Evidence Transfer Memo" templates in the SIB Go Package. **(T-1)** The SIB will hand over privileged evidence not uploaded in AFSAS to the Convening Authority's safety staff. **(T-1)** The Convening Authority safety staff will maintain the privileged evidence until AFSEC/SES publishes the Memorandum of Final Evaluation. **(T-1)** For detailed transfer guidance, reference the Evidence Transfer Flow Chart in the Templates and Guidance of the SIB Go Package.

5.8.2. The SIB president or IO may keep an electronic copy, appropriately protecting any safety privileged information, of all working files, exhibits, etc. in the event any material is needed between when the SIB disbands, and the final message is accepted by AFSEC. The SIB president or IO will destroy all investigation copies after the Convening Authority accepts the investigation results and final message acceptance by AFSEC. **(T-1)**

5.8.3. Delete all network files, folders, e-mail, and backup copies used by the SIB once the investigation is complete.

5.8.4. The SIB will ensure the host wing/delta-supplied computers used by the SIB are reformatted immediately upon conclusion of the investigation, release by the Convening Authority, and return to primary duties. **(T-2)**

5.8.5. The SIB will complete the Mishap Quality Control Checklist and upload it as an exhibit in AFSAS in the Investigation Products exhibit group. **(T-1)**

Chapter 6

SAFETY REPORTS

6.1. Safety Reports. This chapter augments guidance in DAFI 91-204, providing additional content and formatting requirements uniquely applicable to space safety investigations.

6.2. Final Message. The final message is the culmination of the investigation. It provides a complete picture of what happened and why, based on the weight of evidence, professional knowledge, and good judgment of the SIB. SIBs will not refer to exhibits in the final message. **(T-1)** The final message should be detailed enough so the reader can deduce what transpired without any additional information.

6.3. AFSAS Report Requirements. AFSAS provides SIBs with a step-by-step process for entering relevant information and a framework for delivering consistent and quality report requirements. AFSAS will combine appropriate details to create the narrative and streamline the entry process for investigators. The safety report narrative includes the following sections: Sequence of Events, Background Information, Factors, Investigation Conclusions, Acronyms, Documents Reviewed, and Referenced Reports. For directions on completing the narrative, see the SIB Go Package found on AFSAS, and the following additional information:

6.3.1. The Sequence of Event section is a chronological narrative of significant events or actions that led to the mishap. This section summarizes the steps in the mishap sequence until the damage or injury occurred. The SIB will not include analysis in this section. **(T-1)**

6.3.2. The Background Information section includes three sub-sections: General, Object, and Person Background Information.

6.3.2.1. SIBs must use the General Background Information section to provide context and understanding of the systems, processes, or unusual organizations involved in the mishap. **(T-1)** SIBs will not include systems, processes, or organizations not discussed in factors. **(T-1)** SIBs may add photographs, diagrams, and publication screenshots if they aid in describing systems, processes, or organizations. SIBs will not include analysis in this section. **(T-1)**

6.3.2.2. In the Object Background Information section, SIBs will include details about the mishap spacecraft or equipment that caused damage or injury in the mishap. **(T-1)** SIBs will not include analysis in this section. **(T-1)** If relevant for the mishap spacecraft, include the date(s) of the last significant action(s) performed (e.g., software modification, state health actions). **(T-1)** For mishap space system components (e.g., power system, radar or ground equipment), include where the DAF accepted or received the equipment and any significant maintenance. **(T-1)** Including every component that received damage in the Object Background Information section (i.e., damaged parts resulting from original failure) is unnecessary.

6.3.2.3. In the Person Background Information section, SIBs will provide background information on personnel who were factors in the mishap. **(T-1)** SIBs will not include analysis in this section. **(T-1)** If relevant, the SIB will include duty history, significant training, and skill level. **(T-1)**

6.3.3. The Factors section includes factors, causal factors, and non-factors worthy of discussion. See DAFI 91-204 for guidance on writing factors, causal factors, and non-factors worthy of discussion.

6.3.4. The Investigation Conclusions section summarizes why the mishap occurred. SIBs will not have detailed information in this section. **(T-1)** Detailed information is written in the causal factors.

6.3.5. SIBs will include a list of all acronyms used in the narrative in the Acronyms section. **(T-1)**

6.3.6. SIBs will include any other AFSAS reports used during the investigation in the Referenced Reports section. **(T-1)**

6.4. DoD Human Factors Analysis and Classification System (HFACS). Mishaps are rarely attributable to a single cause but are often the result of a series of errors. HFACS aids investigators in identifying the underlying organizational root causes of hazardous conditions to develop more effective risk controls. HFACS classifies human errors into a structured language to understand human contributions to mishaps and explain linkages between individual acts and organizational influences to provide actionable recommendations and improve mishap prevention.

6.4.1. SIBs will input HFACS associated with factors or causal factors in AFSAS. **(T-0)** HFACS will be associated with relevant person(s) or at the event level. **(T-0)**

6.4.2. SIBs should reference the SIB Go Package for the latest HFACS guide. Investigators should consider Human Factors throughout the investigation. Subsequently, codes are applied after the SIB analysis is complete.

6.5. Managing Exhibits. Guidance on a complete list of exhibit groups is provided in DAFI 91-204. For on-duty Class A and B mishaps, most exhibit groups are required. SIBs will upload exhibits that support their analysis and conclusions. **(T-1)** In addition to the requirements found in DAFI 91-204, the following additional guidance is provided:

6.5.1. When uploading files as exhibits to AFSAS, a single .pdf is preferred, multiple .pdfs are acceptable, and when necessary, multiple file types are permissible. SIBs will combine files to include a table of contents to aid reviewers. **(T-1)** SIBs will not upload .zip files as exhibits. **(T-1)**

6.5.2. If a file contains privileged safety information, but there is no place to add the privilege safety information statement, SIBs will create a .pdf file, PowerPoint®, slide(s), or another preferred type of document and add the privilege safety information statement. **(T-1)**

6.5.3. SIBs will not redact any portion of documents uploaded. **(T-1)**

6.5.4. Diagrams. SIBs should include diagrams that increase understanding by showing equipment layout, damage, impact areas, etc. Ensure diagrams are self-explanatory and indicate direction with a northward pointing arrow. If it is practical to do so, indicate the scale. Use Civil Engineering plots, aerial photographs, topographical maps, etc. Do not include diagrams depicting the location of human remains in the "Diagrams" exhibit group. **(T-1)** If they support the SIB's analysis and conclusions, upload such diagrams in the "Medical Information" exhibit group. **(T-1)**

6.5.5. Evidence Transfer Documents**.** SIBs will include either the "Evidence Transfer Memo – No AIB" or the "Evidence Transfer Memo" signed by the BP and receiving agent, indicating the transfer of evidence to any legal board, legal office, safety office, owning organization, etc. **(T-1)**

6.5.6. Guidance, Official**.** SIBs will include highlighted portions of official guidance (DAFIs, DAFMANs, technical orders, etc.) discussed in the safety report narrative, recommendations, or Other Recommendations of Significance. **(T-1)**

6.5.7. Guidance, Unofficial**.** Unofficial guidance includes any unauthorized source not approved by the DAF. If discussed in the safety report, SIBs will upload unofficial guidance (that has not gone through a coordination, review, approval and publication process) such as unit-unique checklists, "how to" guides, individual or unit-created maintenance files, etc., if they support the SIBs analysis and conclusions. **(T-1)**

6.5.8. Investigation Products. Investigation products may include risk acceptance documents, policy letters, etc., that influenced factors in the mishap. If they support the SIB's analysis and conclusions, SIBs will upload these products if they support their analysis. **(T-1)** Include files exclusive to the exhibit group and not included in other exhibit groups. **(T-1)** SIBs will not include files that support mishap cost information. **(T-1)** After the report narrative, findings, recommendations, and HFACS are completed and entered in AFSAS, and all exhibits are completed and uploaded in AFSAS, SIBs will upload a completed Quality Control Checklist to this exhibit group. **(T-1)**

6.5.9. Maintenance and Equipment Records**.** If they support the SIB's analysis and conclusions, SIBs will include records that were in place during the mishap. **(T-1)** Other than laboratory results of equipment tested after the mishap, SIBs will not include records accomplished after the mishap (e.g., corrective actions taken). **(T-1)** SIBs will not include files describing post-mishap repair procedures or requests for repair procedures. **(T-1)** SIBs will not include files that support mishap cost information. **(T-1)**

6.5.10. NDAs. SIBs will include all signed NDAs the ISB or SIB signed or collected. **(T-1)** These include "SIB Member Guidance and NDA," "Non-Disclosure Agreement-Safety Investigation-DoD Personnel, and "Non-Disclosure Agreement-Safety Investigation-Contractor Rep" agreements. It is preferred that the SIB combine all NDAs into a single document. However, the SIB may upload them individually or in groups.

6.5.11. Parametric Data. If uploading reports containing a subset of parametric data specifically requested by the SIB (e.g., selected parameters from the mishap spacecraft), select the attribute "reflects investigative deliberations" and ensure the report contains privileged safety information markings. **(T-1)**

6.5.12. Photographs. If they support the SIB's analysis and conclusions, SIBs will include photographs that increase understanding by showing equipment layout, damage, impact areas, etc. **(T-1)** Combining privileged photographs into a single document may aid the SIB in adding privileged safety information labels more easily. Minimize each photograph file size to aid in combining photographs into a single document to upload into AFSAS.

6.5.12.1. SIBs will not upload every photograph they receive. **(T-1)** Only upload enough photographs to aid the reader in understanding the SIB's analysis and conclusions. **(T-1)**

6.5.12.2. Due to their sensitivity, SIBs will only upload photographs of deceased personnel or injuries in the "Medical Information" exhibit group if they support findings or recommendations. **(T-1)** If the SIB determines to include an injury photo to illustrate the mishap, consider using a black-and-white photo.

6.5.12.3. For non-privileged photographs, SIBs will label each image or name each file to aid reviewers. **(T-1)** Labels (or file names) include name or description and, where applicable, the direction the photograph is facing (e.g., debris field looking west). SIBs will not include information in the label that shows analysis in the label (or file name). **(T-1)**

6.5.12.4. For privileged photographs, SIBs will label each image or name each file to aid reviewers. **(T-1)** Labels (or file names) include the name or description of the photograph (e.g., damaged solenoid, failed gear bolt). SIBs will mark them as containing privileged safety information and select the attribute "reflects investigative deliberations" in AFSAS. **(T-1)** Photographs staged for analysis are considered privileged, and SIBs will mark them as such. **(T-1)** Staged photos are used to illustrate possible scenarios described in the SIB's assumptions or witness testimony. Pointing with a finger or other device at a portion of debris does not constitute a staged photograph and does not make it privileged.

6.5.13. Technical and Engineering Reports**.** The following applies to technical and engineering reports. If a SIB primary or secondary member writes one of these reports, the analysis becomes privileged safety information. When uploading a report written by a SIB primary member in AFSAS as an exhibit, select the attribute "reflects Investigative Deliberations."**(T-1)** Reports written outside of the investigative deliberations are not considered privileged information.

6.5.14. Testimony and Statements. Testimony and statements derived from interviews with witnesses or individuals involved, including statements solicited by the SIB or unsolicited statements voluntarily provided by individuals, will support the SIBs analysis and conclusion.

6.5.14.1. Investigators should start the interview transcription process early due to the time-intensive nature of such tasks; delay may set the investigation behind schedule. If necessary, request additional personnel to help with transcription. Avoid using DAF legal office transcriptionists whenever possible to avoid the appearance of a conflict of interest with any legal investigation or other command actions. SIBs must review and verify the accuracy of all transcripts. **(T-1)**

6.5.14.2. Transcribe and include witness statements in the exhibits if referenced in the SIBs analysis. **(T-1)** SIBs will not summarize the witness testimony. **(T-1)** If testimony is not used to support the SIB's analysis, the SIB will not include it as an exhibit, and will destroy the statements as part of the transfer and disposal of evidence process. **(T-1)**

6.5.14.3. SIBs will not upload interview audio or video files as exhibits. **(T-1) Exception:** If a privileged interview (with a promise of confidentiality) is only partially transcribed, SIBs will include the entire audio or video recording in the Testimony and Statements exhibit group. **(T-1)**

6.5.14.4. SIBs will label all transcription files with the interview date and who was interviewed (e.g., 1 Jan 24 Mishap crew commander Interview). **(T-1)**

6.5.14.5. SIBs will upload a signed Privileged or Non-Privileged Witness Agreement with each written statement or transcribed interview. **(T-1)** SIBs will use the most current documentation for

interviews and written statements in the ISB or SIB Go Packages. **(T-1)** Combine the signed witness agreement with the written statement or transcribed interview into a single document is recommended. When uploading a signed privileged witness agreement, SIBs will select the attribute “obtained with a Promise of Confidentiality.” **(T-1)**

6.5.15. Training Records. If they support the SIB's analysis and conclusions, SIBs will include the crew members' training records for spaceflight mishaps with operations crew involvement. **(T-1)** Records include qualifications, evaluations, and any records showing recent upgrades or training. If other personnel are factors in any mishap and their training records support the analysis, SIBs will include them and select the attribute "reflects Investigative Deliberations." **(T-1)**

6.5.16. Transcripts (Not Interviews). Recordings and transcripts are not privileged. If they support the SIB's analysis and conclusions, SIBs will transcribe audio recordings and include them as exhibits if they support their analysis and conclusions. **(T-1)** SIBs will include a key describing who is speaking (e.g., mishap crew commander, satellite vehicle officer, and ground system operator). **(T-1)** Limit the transcript to periods relevant to the mishap sequence of events but indicate breaks in the transcript. SIBs will not summarize, paraphrase, or otherwise alter the recording when transcribing. **(T-1)** **Exception:** SIBs can replace profanity with "[expletive]."

6.5.17. Videos. If it supports the SIB's analysis and conclusions, SIBs will only upload videos relevant to the mishap and aid the reader in understanding the analysis and conclusions. **(T-1)** Staged videos for SIB analysis are considered privileged. **(T-1)**

6.5.18. Terrestrial weather or space weather. If they support the SIB's analysis and conclusions, SIBs will include weather briefings, actual weather observations, or other weather products. **(T-1)**

6.5.19. Cyber Domain Data. If it supports the SIB’s analysis and conclusions, SIBs will upload relevant cyber domain data (packet captures, IP addresses and mapping, and others).

6.6. Exhibit Waivers. SIBs will request exhibit waivers when there is no data for a required exhibit. **(T-1)** SIBs will use AFSAS to request and route exhibit waivers through the Convening Authority to AFSEC/SES. **(T-1)**

6.7. Safety Report Authentication. Primary members authenticate the safety report by concurring or non-concurring in AFSAS. If a primary member non-concurs in whole or in part with the safety report, the member will submit a Minority Report in AFSAS. **(T-1)** Minority reports will include reasons for disagreement in narrative format, and list suggested findings, causes, and recommendations if different from those contained in the safety report. **(T-1)**

6.8. Classified Reporting. Classified space safety event information will be controlled in accordance with program Security Classification Guidance. SIB’s will contact the Special Security Officer and AFSEC/SES to ensure security requirements are adhered to during the safety investigation and to determine disposition of the classified safety report.

SEAN M. CHOQUETTE
Maj Gen, USAF
Chief of Safety

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

5 USC § 552a, Records maintained on individuals (Privacy Act of 1974)

Title 14, Code of Federal Regulations, Chapter III, *Commercial Space Transportation, Federal Aviation Administration, Department of Transportation*

DoDI 6055.07, *Mishap Notification, Investigation, Reporting, and Record Keeping*, 6 June 2011

AFI 33-322, *Records Management and Information Governance Program*, 23 March 2020

DAFI 91-204, *Safety Investigations and Reports*, 10 March 2021

AFMAN 40-201, *Radioactive Materials Management*, 29 February 2024

DAFMAN 90-161, *Publishing Processes and Procedures*, 18 October 2023

AFPAM 91-206, *Participation in Military or Civil Aircraft or Space Safety Investigations*, 26 April 2019

Prescribe Forms

None

Adopted Forms

DAF Form 847, *Recommendation for Change of Publication*

Abbreviations and Acronyms

AFI—Air Force Instruction

AFMAN—Air Force Manual

AFPAM—Air Force Pamphlet

AFSAS—Air Force Safety Automated System

AFSEC—Air Force Safety Center

AF/SE—Air Force Chief of Safety

AIB—Accident Investigation Board

AMIC—Aircraft Mishap Investigation Course

BP—Board President

CA—Convening Authority

DAF—Department of the Air Force

DAFI—Department of the Air Force Instruction

DAFMAN—Department of the Air Force Manual

DoD—Department of Defense

DoDI—Department of Defense Instruction

DRU—Direct Reporting Unit

FAA—Federal Aviation Administration

FLDCOM—Field Command

FOA—Field Operating Agency

HFACS—Human Factors Analysis and Classification System

IO—Investigating Officer

ISB—Interim Safety Board

MAJCOM—Major Command

MINA—Mishap Investigation Non-Aviation Course

MOA—Memorandum of Agreement

MOFE—Memorandum of Final Agreement

MOU—Memorandum of Understanding

NASA—National Aeronautics and Space Administration

NDA—Non-disclosure Agreement

NTSB—National Transportation Safety Board

No AIB—No Accident Investigation Board

ODMSP—Orbital Debris Mitigation Standard Practices

OPR—Office of Primary Responsibility

ORS—Other Recommendations of Significance

QC—Quality Control

RAM—Radioactive Material(s)

RDT&E—Research and Development Test and Evaluation

SCA—Satellite Control Authority

SIB—Safety Investigation Board

SIO—Single Investigating Officer

SMIC—Space Mishap Investigation Course

Office Symbols

AFSEC/SES—Space Safety Division, Air Force Safety Center

FAA/AST—Office of Commercial Space Transportation, Federal Aviation Administration

SpaceWOC—Space Weather Operations Center, 2d Weather Squadron, 557th Weather Wing

Terms

Anomaly—An unexpected or unplanned condition or event affecting the space, ground or communications segment that does not meet system performance parameters.

Anomaly Resolution Process—Process used to resolve a spacecraft anomaly. The operating/testing organization (squadron or equivalent level) typically conduct this process to resolve a spacecraft anomaly.

Capability Development Document—The Capability Development Document specifies the operational requirements for the system that will deliver the capability that meets the operational performance criteria specified in the Initial Capabilities Document. It outlines militarily useful increments of capability with its own set of attributes and development performance values (i.e., thresholds and objectives).

Class A Mishap—See DoDI 6055.07.

Class B Mishap—See DoDI 6055.07.

Class C Mishap—See DoDI 6055.07.

Class D Mishap—See DoDI 6055.07.

Class E Mishap—See DAFI 91-204.

Collision Avoidance—A process to prevent collisions between on-orbit tracked objects or to prevent collisions between on-orbit tracked objects and launched vehicles (including spent stages)/payloads by determining and implementing courses of action through careful analysis of validated conjunction assessments and satellite health and mission requirements. The process includes establishing launch wait periods in either the launch window or spacecraft maneuvering using validated conjunction assessments and accounts for uncertainties in spatial dispersions, arrival time of orbiting objects and/or the launch vehicle/payload, and modeling accuracy.

Convening Authority—The individual who has the authority to order a safety investigation. For additional guidance refer to DAFI 91-204.

Event—A broad term used to describe an occurrence, a series of occurrences, or a condition which has implications for the safety community. Events include mishap, nuclear surety, incident, hazard, and safety study.

Experiment—A scientific, technological, or developmental investigation or test. An experiment may include several instrument packages from the same or different sponsoring agencies.

FAA-licensed—Launch and reentry operations, the operation of launch and reentry sites, and safety element approvals authorized by FAA/AST under the FAA's authority to regulate commercial space transportation Title 14, Code of Federal Regulations, Chapter III, *Commercial Space Transportation, Federal Aviation Administration, Department of Transportation*, current edition.

Ground-Based Space System—Ground-based space systems include unique space support equipment as well as space-related systems that do not directly support launch operations or on-orbit satellite operations. Examples include warning, surveillance, and offensive/defensive space control systems such as the Space Surveillance Network, the Rapid Attack Identification,

Detection and Reporting System, and the Integrated Tactical Warning and Attack Assessment system.

Hazard—Any real or potential condition that can cause injury, damage, or occupational illness.

Incident—A planned or unplanned occurrence or series of occurrences resulting in injury or damage that does not meet Mishap or Nuclear Surety reporting criteria.

Investigator-In-Charge—A person charged, based on his or her qualification, with the responsibility for the organization, conduct, and control of an investigation.

Launch Vehicle—Any means of transportation used to place an object into Earth orbit or deep space, including ballistic missiles.

Mission Capability—This term encompasses the purpose and functions of the space system (e.g., sensors, transponders, etc.) throughout its intended system mean mission duration (e.g., the expected life of the space vehicle).

Mission Capability Loss—Temporary or permanent loss of the purpose and/or function of the space system at any point during its intended system mean mission duration.

Operational Capability Requirements Document—A document that specifies the operational requirements for a system.

Permanent Loss—The permanent loss of a space system to the extent that one or more of its purposes, functions, or components have become inoperable and unrepairable. The system has come to the end of its life, regardless of its projected lifespan.

Satellite Control Authority (SCA)—The authority to plan, schedule, and perform satellite commanding.

Space Anomaly—An on-orbit malfunction of a space system, or a deviation from what is consistent with normal operations. May utilize the anomaly resolution process when initially investigating an anomaly. A space anomaly may or may not result in damage, injury, or mission impact that reaches the level of a reportable mishap. A space anomaly may be a reportable event.

Space Systems—Any system supporting space operations or support. Space system is a generic term encompassing all ground, space and link segment systems and their components. All the devices and organizations forming the space network are space systems. These consist of: launch vehicles; launch ranges; launch and range support equipment and systems; spacecraft; ground and airborne stations; and data links among spacecraft, mission, and user terminals.

Space Vehicle—A vehicle which orbits or travels beyond the earth's atmosphere or a system for lifting other space vehicles into orbit. Examples of space vehicles include boosters (i.e., launch vehicles, etc.), spacecraft (e.g., satellites, orbiters, payloads, etc.) and reusable spacecraft. Do not consider Intercontinental Ballistic Missiles space vehicles.

Spacecraft—A space vehicle designed to operate in space and launched by a booster. The term includes satellites, orbiters and payloads, and their associated subsystems. This includes manned objects in space. This term can also refer to the object while it is still on the ground. This does NOT include launch vehicle components separated from the spacecraft.

Technical Expert—An individual authorized and qualified to investigate a safety occurrence for a specific spacecraft, system, or process for which they possess unique knowledge or skills. Examples include government and contractor engineers, investigators, and equipment specialists.

United States Government Orbital Debris Standard Mitigation Practices—A set of technical guidelines established to help reduce the number of orbital debris in near-Earth space. The ODMSP apply to all phases of a spacecraft program, from development to disposal. The goal is to limit the creation of new debris, while also minimizing the time that existing debris can remain in orbit.

Attachment 2

ORGANIZATION CONTACT INFORMATION

Table A2.1. AFSEC Contact Information.

Organization	Address	Contact Information
AFSEC/SES (Space Safety) Technical Assistance	9700 G Avenue SE Kirtland AFB NM 87117-5670	DSN 246-6788 or (505)-846-6788 Email: AFSEC/SES Mailbox: AFSEC.SES@us.af.mil